ANKUR SHARMA, PhD

Senior ML & Agentic AI Engineer | Bioinformatics | Cloud Architecture

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PROFESSIONAL SUMMARY

Senior ML and Agentic AI Engineer with 8+ years shipping production systems across clinical genomics, cloud infrastructure, and machine learning. Built multi-agent systems that cut mean time to resolution by 67%, cloud pipelines processing 6,000+ genomic samples at 40% lower compute cost, and open-source agentic AI tooling for variant interpretation. PhD (NTU Singapore). Open to relevant opportunities globally — across industry, research, and startups.

TECHNICAL SKILLS

Agentic AI & LLMs: LangGraph, LangChain, Claude/Anthropic API, OpenAI GPT-4, multi-agent orchestration, RAG, tool-calling/function agents, prompt engineering, LLM routing, Pydantic v2

Machine Learning & Data Science: Python, scikit-learn, PyTorch, Pandas, NumPy, SciPy, statistical modelling, survival analysis (Cox PH), feature engineering, MLflow, Seaborn/Matplotlib

Cloud & Infrastructure: AWS (EC2, S3, Lambda, Batch, Step Functions, DynamoDB, RDS), Docker, Nextflow (DSL2), CI/CD (GitHub Actions), IaC (CloudFormation/CDK), Kubernetes

Bioinformatics & Genomics: NGS (WGS, WES, RNA-seq, ChIP-seq, WGMS, single-cell, spatial), DRAGEN, GATK, samtools, bcftools, pysam, variant calling & interpretation, ACMG/AMP, R/Bioconductor, Snakemake

Regulatory & Clinical: Design Verification Testing (DVT), clinical validation (sensitivity, specificity, LOD), FDA compliance, cGMP, SOPs, HIPAA

OPEN-SOURCE PROJECTS

Reviewer2 — Autonomous ACMG Variant Second Reviewer github.com/ankurgenomics/Reviewer2

4-node LangGraph pipeline (normalise → fetch_evidence → score_acmg → detect_conflicts) for autonomous ACMG/AMP second review. Takes a variant and a proposed clinical classification, re-derives the call independently from evidence, and flags disagreements with the human call. Pydantic v2 validators enforce evidence grounding at the model level: no criterion fires without a source sentence. Exposed as a FastMCP server so any LLM agent speaking Model Context Protocol can call it as a tool. Provider-agnostic LLM layer: Ollama, Anthropic, OpenAI, Gemini. Evaluated against expert-panel ClinVar (ClinGen VCEP) classifications unseen during development: 86% action-band concordance on in-scope cases. 21 tests, CI/CD via GitHub Actions.

Stack: LangGraph, FastMCP (Model Context Protocol), Pydantic v2, Ollama / Anthropic / OpenAI / Gemini, uv, ruff, mypy, MIT

outbreak-agent — Agentic Outbreak Triage System github.com/ankurgenomics/outbreak-agent

4-node LangGraph state machine for real-time outbreak triage. Built around the April 2026 MV Hondius hantavirus scenario: genomic_node (clade + mutation detection) → linkage_node (contact cluster + transmission mode) → risk_node (composite score 0–100 → CRITICAL tier) → critic_node (self-correction loop, max 3 iterations). Produces matplotlib risk dashboard (PNG) and A4 PDF triage report via ReportLab. 33 tests across 3 layers (unit, graph integration, smoke). No API key required for CI.

Stack: LangGraph 0.6, LangChain, matplotlib, ReportLab, pytest, Apache 2.0

agentic-genomics — GenomicsCopilot github.com/ankurgenomics/agentic-genomics

LangGraph agent for explainable variant interpretation. 7-node pipeline: VCF ingest → MyVariant.info annotation (gnomAD, ClinVar, SpliceAI) → frequency filter → Phrank-style HPO semantic similarity scoring → ACMG-lite classification (PVS1 with LoF-intolerance gating, Richards 2015 Table 5 combining rules) → LLM synthesiser → LLM critic that fact-checks claims against evidence JSON. Every run emits a machine-readable reasoning trace. 51 tests, 81% coverage, CI/CD via GitHub Actions.

Stack: LangGraph, Claude API, Pydantic v2, pysam, Streamlit, Typer CLI, pytest, ruff

genomics-skills — Agent-Callable Genomics Skill Library github.com/ankurgenomics/genomics-skills

Modular library of 8 pure-Python skills for downstream genomics analysis: pan-cancer expression profiling (TCGA, 9,479 samples), Kaplan-Meier survival analysis (Cox PH regression), GO/KEGG pathway enrichment, PubMed literature search with NLP digest, protein variant mapping (AlphaFold/PDB), 3D structure viewing, and publication-quality volcano plots. LLM-powered skill routing via Claude Haiku. Parquet caching for instant repeat queries.

Stack: Python, Claude API (LLM routing), cBioPortal/MyVariant/NCBI/PDB REST APIs, Pandas, Matplotlib, pytest (14 tests)

gwas_nf + gwas_bridge.py — Population GWAS → Clinical Interpretation github.com/ankurgenomics/gwas_nf

End-to-end REGENIE-based GWAS Nextflow pipeline for the TEMUS multi-ethnic cohort (4 groups, 10,000 samples, 100,000 variants, 13 phenotypes). Includes **gwas_bridge.py —** a bridge script that extracts genome-wide significant hits ($p < 5e-8$) from `.regenie.filtered.gz` tophit files and formats them as input for GenomicsCopilot's 7-node LangGraph interpretation pipeline. Closes the population discovery → clinical interpretation loop in a single command.

Stack: Nextflow DSL2, REGENIE, AWS Batch, Python, R/ggplot2, Docker

PROFESSIONAL EXPERIENCE

Senior Scientist, Bioinformatics Jul 2025 – Present

Illumina · Singapore

- Lead **Design Verification Testing (DVT)** and regulatory validation for TSO500, NIPT16, and VeriSeq diagnostic products — ensuring FDA compliance and clinical-deployment quality.
- Designed clinical concordance studies measuring sensitivity, specificity, accuracy, LOD, and reproducibility for regulatory submissions.
- Optimised TruSight Oncology 500 assay workflows for comprehensive genomic profiling in oncology; contributed to NIPT16 prenatal testing validation.

Senior Scientist — Bioinformatics & Cloud Jan 2022 – Jul 2025

Mirxes · Singapore

- Built production-grade cloud data-analysis pipelines: **40% ↓ compute cost, 50% ↓ storage footprint, 30% ↓ turnaround time**.
- Processed **6,000+ genomic samples** with standardised Nextflow workflows and rigorous QC on AWS.
- Architected AWS infrastructure integrating **15 services** (Batch, Lambda, Step Functions, S3, DynamoDB) to manage **400 TB** of genomic data.
- Led team of 5 scientists delivering clinical diagnostic assay workflows across cancer types — 33% ↓ analysis time.
- Technical lead for the **Singapore National Precision Medicine** project; identified SGD 0.5M in business opportunities.

Scientist, Assay Development Jul 2021 – Dec 2021

Vela Diagnostics · Singapore

- Implemented automated verification procedures and QC protocols; maintained regulatory documentation (NCR, CAPA, DR).
- Developed testing framework that cut non-conformance reports by 30% and lifted product quality by 15%.

PhD Research Fellow 2016 – 2021

Nanyang Technological University · Singapore

- Multi-omic study (RNA-seq, ChIP-seq, Hi-C) of age-dependent transcriptional and epigenetic alterations in mouse hepatocytes.
- Built reproducible NGS analysis pipelines for transcriptome, histone modifications, and 3D chromatin data.
- Head of NTU 3-Minute Thesis team (23 members, Nanyang Awards); led operations for TEDxNTU (78-person team, 1,500+ audience).

Technical Supervisor 2011 – 2013

Zydus Cadila Healthcare Ltd. · Ahmedabad, India

- Developed process-validation and QC pipelines for 13 pharmaceutical products under cGMP guidelines — 30% ↑ efficiency, 10% ↓ processing time.

EDUCATION

PhD — Nanyang Technological University, Singapore (2016–2021)

Dissertation: Age-dependent transcriptional and epigenetic alterations in mouse hepatocytes · DOI: 10.32657/10356/155390

M.Tech Biotechnology (2013–2015)

B.Tech in Biotechnology — Uttar Pradesh Technical University, India (2007–2011)

PUBLICATIONS

Sharma, A. (2021). *Age-dependent transcriptional and epigenetic alterations in mouse hepatocytes.* PhD Dissertation, NTU Singapore. DOI

Sharma, A. et al. (2019). Significance of hepatocyte polyploidization in liver physiology and pathology. *Cell Symposia, Chicago.*

Kumari, M., Taritla, S., **Sharma, A.**, & Jayabaskaran, C. (2018). Antiproliferative and antioxidative bioactive compounds in marine-derived endophytic fungus. *Frontiers in Microbiology.*